We also recommend that you increase reviewer friendliness by emphasizing key words that identify components you know reviewers will need to find to review your proposal. We will tell you which words to emphasize later in this chapter. *Judicious use* of *subtle* emphasis (italics and underlining) is the key; overuse causes such emphasis to lose its effect and can even become an aggravation to reviewers.

EXPANSION OF YOUR BULLETED OUTLINE INTO SENTENCES

Once you are satisfied with the bulleted outline you created, the next step is to expand it into your first draft of the Overview subsection.

Introductory Paragraph

Opening Sentence. Expand the bullet that you wrote for your opening sentence. Make sure that it reads well and flows easily into the presentation of what is currently known. As described in the previous chapter, make sure it contains a "hook," an indication of the proposal topic, and some conceptual linkage to the RFA or program you're targeting. Make sure that the sentence is arresting and, as such, will immediately attract the reviewer's interest and attention.

Current Knowledge. Working from the bullets that reflect current knowledge, write 3 to 5 complete sentences that convey what is known about your subject area. This series of sentences should quickly and concisely convey what the scientific landscape is about the specific puzzle or problem you are addressing in this proposal. To craft these sentences, consider the following questions: What do we already know, or what is already suggested about this puzzle? What have others established? In trying to address or solve the problem, what has already been tried; what worked, what didn't, or what did but only partially? Remember that your own data can count as current knowledge. Cite only the most important references. Here are a few tips for determining whether the reference would be considered a key reference. First, if the sentence would be followed by two or more citations, then consider that it is generally known. Second, if the sentence would not be believed without citation, include the citation. Third, if the work is from your own group, include the citation especially if you are a young investigator.

Gap in the Knowledge Base/Lack of Something. Next, expand the bullet that you wrote to delineate the gap in the knowledge base or the lack of something that will drive the application. Often, what works well here is for the first part of the statement to provide a quick summary of the "current knowns" you just explained, with the second part of the statement providing the specific gap or unknown. For example:

- "While these findings collectively suggest X, Y remains unknown/unspecified."
- "Although these studies have provided a general sense of X, there is a knowledge gap regarding the precise mechanisms by which Y occurs."
- "This knowledge has facilitated X but has not translated to Y because Z remains unsubstantiated."

Statement of Need/Consequences of Not Meeting the Need. These bullets should be expanded into two sentences. The first will frame the need, which will of course be related to the gap in knowledge (or lack of something). For the first sentence, state clearly that there is an urgent or critical need. "There is, therefore, a *critical need* to..." Note that *"critical need"* is underlined and italicized. If publications of other investigators are used to support existence of the need, cite that work. Next expand the bullet on why the continued existence of that need is a problem relevant to the USDA.

Be sure that your potential consequences statement is congruent with the USDA's own description of their priorities. You can use key concepts or words from the RFA description and/or the program priority description to craft this sentence. For example, imagine you are an investigator focused on improvements to the ventilation in swine housing for gestating sows. You might be submitting a proposal in response to the AFRI Foundational RFA for FY 2021 and FY2022. Specifically, your work falls under Program Area Priority Code A1251: "Welfare and Well-being of Agricultural Animals." In the RFA, part of the description for this Program Priority reads:

"Evaluation of current animal agriculture production practices and/or development of new or enhanced management approaches that safeguard both animal welfare and sustainable production efficiency, including, but not limited to:

- Alternatives or improvements for painful management procedures; euthanasia and slaughter methods to decrease pain and distress; handling and transportation to decrease injury and distress (including heat stress)"

Using the concept of heat stress from the priority description, you could craft a potential consequences statement along these lines:

"Without this knowledge, the development of ventilation standards for gestating sows in group pens will remain impossible, risking harmful heat stress for sows, and suboptimal swine performance for producers."

What, Why, Who Paragraph

Long-Term Goal. Expand the bullet for your long-term research goal. It should clearly encompass the gap or unmet need that was delineated in the first paragraph. Do so by completing the following: "Our [My] *long-term goal* is to ...." Note that the words "long-term goal" have been underlined and italicized to make them easier for reviewers to find. Note, also, that there is a hyphen between "long" and "term." It is required because those two words created a compound adjective that modifies "goal."

This is another place that you can potentially incorporate key concepts or words from the RFA or program description. For example, the following long-term goal statement echoes key concepts from the Program Priority description for A1201: Animal Breeding and Functional Annotation of Genomes in the FY 2021-2022 AFRI Foundational RFA:

"Our *long-term goal* is to directly inform the development of national breeding strategies to reduce abiotic stresses related to X in the context of Y."

In this case, the shared goal with A1201 is "development of national breeding strategies to reduce abiotic stresses," and X and Y encapsulate your niche area.

Overall Objective. Now, expand the bullet you wrote for the overall objective into a single sentence that conveys what you want to produce with this research project. Make sure that what you write here links back to the gap in the knowledge base or unmet need that you presented in the first paragraph; the objective of any grant application must be to either fill that gap or meet that need. Label this sentence by beginning it with something like, "The *overall objective* of this application is to…" example: "The *overall objective* of this application is to determine which structural characteristics of the β protein allow it to function in the bovine IFN-γ receptor complex."

In crafting your overall objective, remember to avoid indeterminate verbs such as "to study," or "to investigate," or "to examine." Instead, try answering the question of "why." Why are you examining X? Why are you comparing A and B? Usually, the answer is that you are trying to determine, identify, establish, or confirm something. Try using those words in your objective instead. For example:

- "Our *overall objective* is to identify barriers to utilization of insect protection netting among producers of crop X."
- "Our *overall objective* is to establish the cost effectiveness of strategy X for doing Y."
- "Our *overall objective* is to determine the processing conditions that will optimize A."

When you have completed this sentence, make certain that it links back to the gap/lack and statement of need components and that it relates clearly to your long-term goal.

Central Hypothesis and How Formulated. This component will be included if you are writing either a hypothesis-driven application or a "hybrid" – one that has both hypothesis-containing and need-driven aims. The expanded bullet should convey your overarching prediction regarding how the overall objective will be attained. Ideally, it should have readily identifiable parts that will be used to set up the specific objectives/aims. Be certain that what you have proposed is testable, and does not contain conditional words like might, may, likely, or can. Additionally, make sure that you label this component so that it can be found easily by reviewers.

For example, "The *central hypothesis* for the proposed research is that function of the β chain in the bovine receptor complex is dependent on association of its omega helix with a complementary structure on the α chain, resulting in the creation of a functional receptor."

Another example might be, "Our *central hypothesis* is that a user-friendly, smartphone-based integrated water management system will decrease overall water usage and fertilizer runoff, while improving crop yields and costs."

> **TIP:** The first 3 components in this paragraph create a focusing progression, which the reviewer must perceive. To help ensure that they make that connection, the long-term goal, overall objective, and central hypothesis should be presented without interposed explanatory information. These three components should be juxtaposed, so that there is a seamless flow of logic from the broadest (the long-term goal) to the narrowest focus (the central hypothesis).

Follow the statement of your central hypothesis with another sentence that expands the bullet that you wrote to describe how your hypothesis was formulated, i.e., why it represents the most attractive avenue of research to pursue first. Remember that this is the second place that you can incorporate your earlier findings into the overall argument or story. For example, "We have formulated this hypothesis based on strong preliminary findings, which suggest that _____." If the work of others is used in addition to your own preliminary data to support formulation of your central hypothesis, the relevant publications should be cited. Other examples:

- "This hypothesis is grounded in our preliminary data indicating Q."
- "This hypothesis is based on our prior data suggesting X, which echoes and extends that of Archer and Katzenberg (2019)."

Now write your own central hypothesis, then expand the bullet(s) that you wrote to describe how your hypothesis was formulated, i.e., why it has been chosen as your "best bet."

Well-Prepared and Research Environment. Next, using the bullets that you wrote earlier, write 1 to 2 complete sentences that summarize why you and your colleagues have the competitive edge in this area of research. Remember that if no one feature makes your program stand out, you will want to emphasize the constellation of your research team's strengths. This statement also helps meet the statement in most USDA application instructions to describe, where applicable, in the *Introduction*, "reasons for performing the work at the proposed institution."

Regardless of what you write here, you must be very careful how you construct this component. The last thing that you want to do is inadvertently convey that you are overly impressed with your own credentials/accomplishments. An acceptable set of sentences might be something like, "We are particularly *well prepared* to undertake the proposed research because our team consists of a protein chemist, a membrane biophysicist, and a molecular biologist, each with long-standing research interests in signaling by bovine IFN-γ. Additionally, we have unique access to a research herd of susceptible cattle and equipment X, making our environment especially conducive to successful completion of the proposed research." Now, write your comparable sentence(s) in your draft document.

Specific Objectives "Paragraph"

Begin paragraph three with a sentence something like, "We propose [two/three] *specific objectives*." For integrated projects, you can also harness this sentence to remind the reader that the work is integrated; for example, "We plan to achieve our overall objective with the following [two/three] integrated research and extension *aims*:" Next, expand the bullets for your specific aims and the working hypothesis for each. After doing so, make certain that each clearly links back to a part of your central hypothesis.

Because the specific objectives are so important, we recommend that you set each off as a bolded, italicized "headline" statement, and that it and its subordinate paragraph be indented one tab. Nothing in the subordinate paragraphs should be bolded or otherwise highlighted, except for "*working hypothesis*" described for the hypothesis-containing specific objectives.

For the extension- or education-related specific objectives, have a bolded headline of what you plan to do. As mentioned before, these aims can be more task-oriented, but try to focus on what you plan to accomplish by the activities. Instead of saying "Conduct seminars for producer boards," or "Create displays for public education in local agriculture"—*what* you plan to do—say something that depicts the "*why*." For example, "Update and increase producer skills in X" or "Improve public knowledge of local agriculture production." This type of specific objective would be followed by a sentence that describes *how* you plan to do it (e.g., conduct seminars for producer boards, or create the displays featuring A and B). This indication of the methods or approach will help the reviewer follow your thinking of what you want to do and how, and increase their confidence that you have a plan to carry out the work.

Payoff Paragraph

Innovation/Creativity/Originality. If you have written a bullet that describes why your research/approach is innovative, expand it into a sentence that will open paragraph four. If you have a credible claim to innovation, be direct about it. Remember to create as much contrast as possible between the "status quo" and what you are proposing to do. For example, "The proposed project is innovative because in doing/considering/addressing X, it departs from the status quo, which typically only involves Y."

Remember, that some RFA instructions (e.g., AFRI Foundational, and Sustainable Agricultural Systems), ask that statements of novelty/originality be placed in *Rationale & Significance*!

Expected Outcomes. Continue by expanding the bullets that present your expected outcomes and why they are important. If you have used an "innovation" sentence to open this paragraph, you might begin presenting your expectations by using a segue, such as: "This approach is expected to yield the following outcomes: identification of X, confirmation of Y..." Alternatively, if you didn't open with a statement about innovation, begin the fourth paragraph with a sentence something like: "At the completion of these studies, our *expectation* is to have identified X, confirmed Y..."

Positive Impact. Conclude the fourth paragraph and your Overview by expanding the bullets that you wrote to summarize the general impact that your research will have. It might be similar to something like, "By collectively establishing how structure of the β protein dictates its role in transmembrane signaling by the bovine IFN-γ receptor complex, these outcomes are intended to inform new pharmacologic strategies and thus have a <u>*positive impact*</u> on bovine health and well-being."

When crafting the positive impact statement, choose your words carefully. Avoid saying that you "hope" or "believe" that your results will have a positive impact, or that your findings "should" have a positive effect. At the same time, avoid too many instances of "will," which can be perceived as being overly confident or over-promising. Instead, choose words that convey reasonable confidence, e.g., "We anticipate/intend/project/expect that these results will have a *positive impact*..."

FINALIZING YOUR *OVERVIEW* SUBSECTION

First Draft of the Overview Part of Your Introduction Section. Revise and refine the sentences until they flow well. Add material, if needed, to link the sentences appropriately and blend them into readable prose. If the compelling flow of logic that you need isn't there, you may have to rewrite the components that don't work. It may even be necessary to go back to the bulleted outline stage. Whatever it takes, invest the effort until this first, Overview subsection of your Introduction works and accomplishes its purpose. When you are done, you should have approximately 1.0 to 1.5 pages. If you have more than that, prune what you have written until you meet that mark. Switching to a smaller font and narrowing the margins are not options, because of USDA rules for formatting. Taking out open lines between paragraphs to stay within 1.5 pages should not be an option, because sacrificing readability will not serve you well.

Concluding Work on the Overview Part of Your Introduction Section. At this point, you should have an Overview that succinctly conveys to your reviewers, in a compelling and exciting way, what's known, what's not known, what you are going to do about it, what your expectations are, and why the expected results will be agriculturally important. If you have time, show your draft to one or more of your colleagues to determine whether it is as clear and compelling a presentation as you think it is. If it still needs work, continue to revise this subsection until you are satisfied. Before continuing with the remainder of the proposal, be sure to have your colleagues read this subsection. As discussed in Chapter 20 on the Pre-Submission Review Committee, internal peer-review of this Overview is a critical step in creating a competitive application.

LEARNING MILESTONES FOR CHAPTER 7:

1. Understand the iterative nature of expanding the bullets for your Overview subsection into complete sentences, and appreciate how their expansion creates a compelling, logical argument for the Overview.
2. Recognize the value of labeling these components with underlining and italics as a reviewer-friendly strategy that helps reviewers know precisely why the related information has been included.
3. Appreciate how making your specific aims stand out with tabs and bolding serves as a visual cue and provides easy recognition by reviewers.
4. Appreciate the importance of citing only key references in this subsection.

Example of a Hypothesis-driven Overview Subsection

This Overview subsection from a USDA/AFRI-funded project is reprinted with the permission of Principal Investigator Dr. Caleb Lemley of Mississippi State University.

MELATONIN-MEDIATED REGULATION OF BOVINE PLACENTAL CIRCADIAN RHYTHMS DURING COMPROMISED PREGNANCIES

AFRI Competitive Grant, PROJ NO: MIS-339090

INTRODUCTION

Overview

Proper fetal nutrition via adequate uteroplacental blood flow is critical to maximize bovine offspring production potential while minimizing pregnancy wastage and calf morbidity and mortality, which are estimated to cost producers over $1 billion annually (USDA-NASS, 2011). Inadequate maternal nutrition leading to a compromised pregnancy increases calf mortality rate by 10% at or near birth, while an additional 20% of calves die prior to weaning (Corah et al., 1975). Common signs of a compromised pregnancy are decreased uteroplacental blood flow and placental nutrient transport, which are both indicators of placental functional capacity (Fowden et al., 2006). Recently our team observed circadian rhythms in bovine uterine blood flow and placental clock gene expression, which were increased from the early morning to afternoon following dietary melatonin supplementation. While our initial findings are intriguing, the therapeutic potential of ameliorating inadequate daytime uterine blood flow with dietary melatonin during a compromised pregnancy remains unknown. Thus, there is a *critical need* to identify the physiological pathways through which melatonin mediates placental rhythms in the context of a compromised pregnancy. In the absence of such information, the substantial economic losses of offspring born from a suboptimal uteroplacental environment will persist.

Our *long-term goal* is to increase the efficiency of livestock animal production by identifying novel pathways that can be used to improve placental functional capacity during a compromised pregnancy. The *overall objective* for this application is to identify melatonin-induced circadian rhythms in placental functional capacity and gene expression in the context of a compromised pregnancy. The *central hypothesis* for the proposed research is that exogenous melatonin during the daytime will ameliorate inadequate placental functional capacity by increasing the daytime amplitude of placental clock genes during maternal nutrient restriction. This hypothesis was formulated based on strong preliminary findings, which indicate that dietary melatonin supplementation in the morning increased daytime uteroplacental blood flow and placental clock gene expression, while maternal nutrient restriction decreased daytime uterine blood flow compared to that observed in adequately fed dams. Our diverse team of investigators is especially *well-prepared* to conduct the proposed work as it consists of research scientists and veterinarians with a long-standing collaborative history and expertise in reproductive biology, large animal theriogenology, and ruminant nutrition. Additionally, our research environment is especially conducive for the successful completion of the work as we have access to the necessary animal handling facilities and large animal surgical suites. We will test the central hypothesis by pursuing the following *specific objectives*:

Specific Objective #1: Identify differences in circadian rhythms in bovine uterine and umbilical blood flow in adequately fed and nutrient-restricted dams supplemented with or without dietary melatonin. Our working hypothesis, based on our aforementioned preliminary data demonstrating opposite responses in daytime uterine blood flow during maternal nutrient restriction versus melatonin supplementation, is that melatonin supplementation during the daytime will mitigate the maternal nutrient restriction-induced decreases in daytime uterine and umbilical blood flow.

Specific Objective #2: Identify differences in circadian rhythms in placental clock gene expression in adequately fed and nutrient-restricted dams supplemented with or without dietary melatonin. Our working hypothesis is that the daytime amplitude of placental clock gene expression and clock-controlled angiogenic factor expression (e.g., VEGF, COX2) in melatonin supplemented dams will be greater than that observed in dams with compromised pregnancies brought about by nutrient restriction. This hypothesis was formulated from our aforementioned preliminary data demonstrating an increase in bovine placental clock gene expression following daytime melatonin supplementation.

At the conclusion of this research, our *expectation* is to have identified melatonin as a regulator of circadian rhythms in uteroplacental blood flow and placental clock genes. Moreover, we will determine the feasibility of an *in vitro* model for examining 24-hour rhythms in placental functional capacity, which is more practical for generating larger data sets related to placental circadian research. These outcomes are expected to have significant *positive impacts* on maximizing ruminant placental efficiency and directly informing the development of superior strategies to mitigate wastage of placental functional capacity.